

Fig. 5: Interfacing of LCD display with an FPGA

TABLE I
Different Commands to the
LCD Instruction Register

Code (Hex)	Command to LCD instruction register
1	Clear display screen
2	Return Home
4	Shift cursor to left
6	Shift cursor to right
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to left
1C	Shift the entire display to right
80	Force cursor to beginning of first line
C0	Force cursor to beginning of second line
38	2 lines and 5×7 matrix

8. Write 0x06, wait for 4.2 milliseconds (shift cursor to right)

9. Write ASCII 'W', wait 4.2 milliseconds

10. Write ASCII 'E', wait 4.2 milliseconds

11. Write ASCII 'L', wait 4.2 milliseconds

12. Write ASCII 'C', wait 4.2 milliseconds

13. Write ASCII 'O', wait 4.2 milliseconds

14. Write ASCII 'M', wait 42 milliseconds

15. Write ASCII 'E', wait 4.2 milliseconds

As shown in the flowchart in Fig. 6, all steps do not require a few milliseconds for execution. However, to

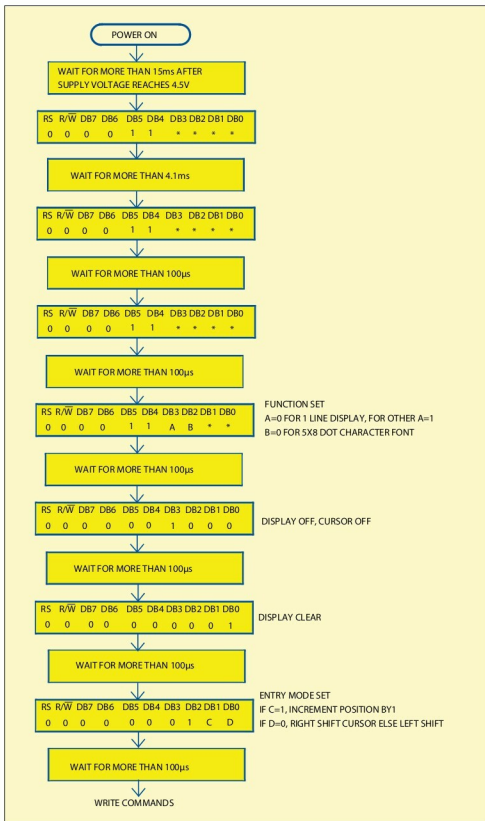


Fig. 6: Flowchart for LCD initialisation

simplify the code, the time for all the steps is kept uniform. In case this is found unsuitable for the design, the designer can modify the timings.

Instead of using delays, one can check the status of the busy flag to see if the LCD is ready to receive information. The busy flag is the MSB of the

data bus and can be read when $R/\bar{W} = 1$, $RS = 0$ and low-to-high pulse is applied to the EN pin. If the busy flag is high, the LCD is busy taking care of the internal operations and will not accept any new information. When the busy flag is low, the LCD is ready to accept new information.